Component Specification Document

Total pages: 38 pages

Test / production New / Replacement	Title New Mimamori ECU examination demand specification document	Specification document number 1SKTS60500

Distribution list

7-SG2	①
7-SG3	①
7-SK7	①
7-SK8	①
7-SKE	①
7-RT2	①
7-SJ6	1
7-SJ8	1
1-N42	1
7-N42	1
7-T6E	1
7-T6D	1
7-XF2	1
7-XF3	1
7-XF4	1
7-X11	1
J-BUS	1
Total	17 #

Names with ○ receive cover only

A. Applicable models, model years, device components
B. Regulation summary and table of contents
C. Summary of changes recorded in order.

Related document numbers

Configuration					
A0		A2		B4	
A1A		A3		B5	
A1		A4	38	Total	38
Control No.					

A. Applicable models

Vehicles with new mimamori ECU installed
-VC30, VF47-6H, VN57, VN60
-VB13-Diesel, VB-13-CNG-LV

B. Regulation summary

This specification document describes the new mimamori EOL data flash (input) process using the factory investigation equipment and summarizes the settings and requirements (including QA table) necessary for inspection.

C. Summary of changes

		(5)	
		(4)	
		(3)	
		(2)	
		(1)	
		0	Aug 18, 2009 6Z-0994

Revision (issue) history

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Related				

[Stamp: Received
Aug 18, 2009
IEC Drawings Dept]

Revision History

Date	Ver	Change details	Notes
July 14, 2009	0	Newly issued	

Table of Contents

1. Purpose 5
2. QA Table 5
2.1 QA Table 5
2.2 Wiring 6
2.3 Pin Layout 6
2.4 EOL Flash Process and Examination Items 7
3. Communication Protocols 8
3.1 EOL Communication Protocols 8
3.2 Specialized Vehicle Equipment/Data Download Communication Protocol 8
4. Related Requirement Specifications Documents 8
5. EOL Input Process/Examination Flow 8
6. EOL Data Flashing Process 9
6.1 Data for Flashing 9
6.2 Flash Process 9
6.2.1 Start Diagnostic Communication 10
6.2.2 Diagnostic Session Start (Programming mode) 10
6.2.3 Security Access (Check security conditions) 11
6.2.4 Flash ECU ID (Flash VIN) 11
6.2.5 Flash EOL data 1 (Flash mimamori data) 12
6.2.6 Flash EOL data 2 (Flash data for preventative maintenance) 14
6.2.7 Diagnostic communication completion 15
7. EOL Data Examination Process 16
7.1 EOL Examination Items 16
7.2 EOL Examination Procedure 16
7.2.1 Start Communication 17
7.2.2 ECU ID Download (VIN download) 17
7.2.3 EOL Data Download 1 (download data for mimamori) 17
7.2.4 EOL Data Download 2 (download data for preventative maintenance) 19
7.2.5 Security Access (ECU lock) 20
7.2.6 Security Access (Check security access) 20
7.2.7 Complete Diagnostic Communication 20
8. Operation Confirmation Examination Processing 21
8.1 Operation Confirmation Preparation 21
8.2 Confirmation Items 21
8.3 Confirmation Procedure 21
8.4 Shared Information Protocol Communication Start Process 22
8.5 DTC Clear/Check (When executed via shared information protocol) 25
8.5.1 DTC Clear 25
8.5.2 DTC Check 26
8.6 DTC Clear/Check (When executed via KWP2000) 27
8.6.1 Diagnostic Communication Start 27
8.6.2 DTC Clear 27
8.6.3 DTC Check 27
8.6.4 Diagnostic Communication Completion 27
8.7 Clear All Mimamori Data (shared information protocol) 28
8.7.1 Clear All Mimamori Data 28

8.7.2 Frequency Data Check 29
8.7.3 DRM/EDR Data Check 30
8.8 Clear All Mimamori Data & Check Process (KWP2000 protocol) 32
8.8.1 Diagnostic Communication Start 32
8.8.2 Clear All Mimamori Data 32
8.8.3 Diagnostic Communication Completion 32
9. EOL Data 33
9.1 EOL Data Definitions for Mimamori Use 33
9.1.1. Installed ECU Data (LID: $20~$3F) 33
9.1.2. Engine Model Data (LID: $40) 34
9.1.3. T/M Model Data (LID:$41) 34
9.1.4. Tire Radius Data (LID:$42) 34
9.1.5. Final Ratio Data (LID:$43) 34
9.1.6. Vehicle Model Data (LID:$44) 34
9.1.7. Electrical Equipment Data (LID:$45) 35
9.1.8. Gear Ratio Data (LID:$47) 35
9.1.9. Vehicle Speed Correction Coefficient (LID:$48) 35
9.1.10 New Mimamori Calibration Data (LID: $50) 35
9.2 Preventative Maintenance EOL Data Definitions 36
9.2.1. Minimum Guaranteed Drive Distance for Engine Oil Life (LID:$52) 36
9.2.2 Drive Distance for Engine Oil Life Warning (LID: $53) 36
9.2.3 Engine Oil Life Prior Warning Drive Distance (LID:$54) 36
9.2.4. Engine Oil Life Warning Total Engine Revolutions (LID: $55) 36
9.2.5. Engine Oil Life Prior Warning Total Revolutions (LID: $56) 36
9.2.6. Engine Oil Life Warning - Insoluble Material (LID: $57) 36
9.2.7. Engine Oil Life Prior Warning - Insoluble Material (LID: $58) 36
9.2.8. Starter Life Warning Number of Starts Data (LID: $59) 36
9.2.9 Exhaust Brake Valve OH Warning Number of Actuations (LID: $5A) 37
9.2.10 ACG OH Warning Drive Distance (LID:$5B) 37
9.2.11 ACG OH Warning Number of Operations (LID:$5C) 37
9.2.12 T/M OH Warning Number of Gear Engagements (LID:$5D) 37
10 DTC Codes 38

1. Purpose

This specification document specifies the EOL flash (input) process and investigation items required for the new mimamori ECU. For the new mimamori ECU, flashing of the EOL data in accordance with the model/spec and VIN number (SPS support) is necessary.
In addition, confirmation that the handling of data between mimamori <--> multi-meter is displayed correctly, the possibility of mistakes remaining in the processes, and deletion of diagnostic and recording data are necessary.
After flashing the EOL data, operate the multi-meter and confirm that the fuel consumption gauge is displaying correctly.
In addition, confirm that normal communication for the special vehicle device/data download is possible, and clear diagnostic data and recording data before shipment from factory.

Electrical Equipment/Control Development Dept Control Design Group 3		
Approved	Reviewed	Created
		Takeda

2. QA Table

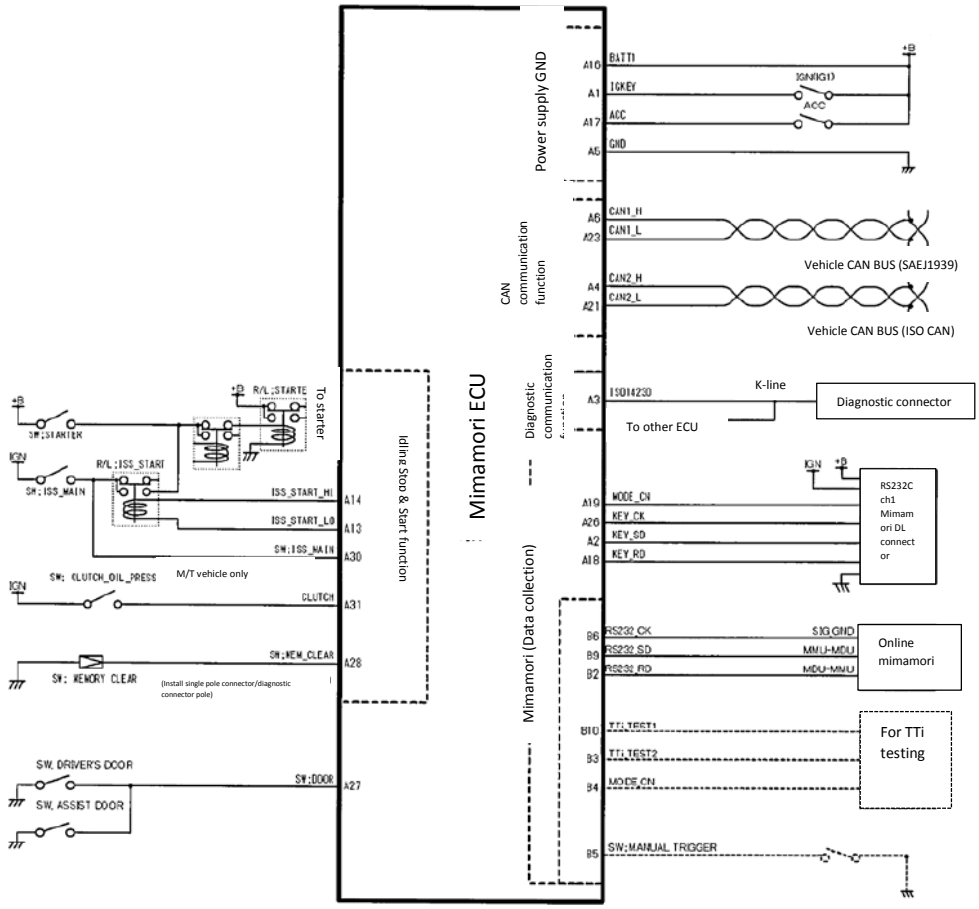
2.1 QA Table

The QA table for the new mimamori ECU is shown below

		Control Unit Mimamori		Change date:	Creation date: 6 July 2009						
No		Device name Function name	Function	Severity *1	Product management point (operations check points)	Related terminals (Refer to appendix)	Effect on upper level systems if design quality is not met	Self diagnostic	Communicating Dept	Detected Y/N	Feedback from communicating dept (exam results)
1	ECU	EOL	Select appropriate operation based on data flashed by EOL	A	EOL data flashed in accordance with vehicle.	-	Mimamori data may not be recorded properly.	Partial	MFR QA		
		Self-diagnostic	Self-diagnostic	A	Self-diagnostic DTC is not raised.	-	Mimamori data may not be recorded properly.	Yes	MFR QA		
		Frequency data recording	Records frequency data in order to understand vehicle usage, etc.	B	Frequency data is cleared before leaving factory. All ECU do not fail after clearing.	-	Vehicle may be shipped out with data accidentally recorded when it was not in a shippable condition, data leaked to market.	No	MFR QA		
		DRM recording	Records control data, DTC, etc. for before & after subject ECU failure. (DRM function)	B	Frequency data is cleared before leaving factory. All ECU do not fail after clearing.	-	Shipped with diagnostic records still on board, data leaked to market.	No	MFR QA		
		EDR recording	Records before & after control data when subject ECU operates and/or when recording conditions are satisfied. (EDR function)	B	Frequency data is cleared before leaving factory. All ECU do not fail after clearing.	-	Shipped with data accidentally recorded in the factory, data leaked to market.	No	MFR QA		
		Communication with multi-meter	Operates fuel consumption gauge. Displays fuel consumption data.	A	Connectors are fully connected. No damage to harness during work process handling. Capable of communicating properly with factory equipment. After flashing EOL, fuel consumption gauge is displayed.	-	Fuel consumption data etc., mimamori <--> multi-meter data handling for multi-display is not displayed.	Partial	MFR QA		
2	Harness	Power supply	Supplies power (B shift, KEY power, ACC power) and GND to ECU/	A	Connectors are fully connected. No damage to harness during work process handling. CAN communication related DTC is not raised.	A1, A5, A16, A17	Mimamori system does not operate, data is not recorded. Fuel consumption cannot be displayed.	No	MFR QA		
		CAN communication	Provides CAN communication to other ECU.	A	Connectors are fully connected. No damage to harness during work process handling. Capable of communicating properly with factory equipment.	A6, A23, A4, A21	Mimamori data may not be properly recorded. Fuel consumption cannot be displayed.	Yes	MFR QA		
		KWP2000 communication	Performs KWP2000 communication with tester/factory equipment, sets the EOL data, and checks snapshot data.	A	Connectors are fully connected. No damage to harness during work process handling. Capable of communicating properly with factory equipment.	A3	EOL cannot be flashed. Service tool (IDSS/TechII) cannot be used.	No	MFR QA		
		Data download RS232C communication (1) Shared information protocol	Provides RS232C communication (shared information protocol) for computer/vehicle devices, downloads accumulated data.	A	No damage to harness during work process handling. Capable of communicating properly with factory equipment.	A2, A18, A19, A26	Cannot download mimamori data.	No	MFR QA		
		Communication with online mimamori RS232C communication (1) Shared information protocol	Provides RS232C communication (shared information protocol) for online mimamori, transmits data.	A	Connectors are fully connected. No damage to harness during work process handling. Capable of communicating properly with online mimamori (For confirmation items, follow online mimamori side instructions) *2	B2, B6, B9	Cannot communicate with online mimamori	No	MFR QA		
		Door SW	Transmits Door SW condition over CAN	B	Connectors are fully connected. No damage to harness during work process handling.	A27	Cannot transmit Door SW condition properly over CAN.	No	MFR QA		

*1: Severity
A: Stops functionality
B: Degrades functionality
C: Reduces marketability
*2: Carry out only when online mimamori is set/installed

2.2 Wiring



For the harness indicated with the dotted line, these are improved additional components after vehicle completion, or spare circuits. These do not need to be implemented in the vehicle harness.

2.3 Pin Layout

Seen from connector fitting face.

Connector A

A16	A15	A14	A13	A12	A11	A10	A9	A8	A7	A6	A5	A4	A3	A2	A1	
	A32	A31	A30	A29	A28	A27	A26	A25	A24	A23	A22	A21	A20	A19	A18	A17

ECU side connector (JAE MX34032NF2)

Connector B

B6	B6	B4	B3	B2	B1	
	B12	B11	B10	B9	B8	B7

ECU side connector (JAE MX34012NF1)

NOTE: Connections do not need to be implemented in the vehicle harness

Pin #	Name	Signal details	Spec	Function
A1	IGKEY	IGN1 Input	SW Input (close to +B)	-
A2	KEY_SD	RS232 ch1 transmission data	TIA232F standard	-
A3	ISO14230	Diagnostic communication terminal	ISO14230-1 standard	-
A4	CAN2_H	Vehicle CAN bus (hi)	ISO11898 standard	-
A5	GND	BCM GND	-	-
A6	CAN1_H	Vehicle CAN bus (hi)	SAE J1939/11 std	-
A7	SW-YOBI(4)	Spare circuit	SW Input (Close to GND or IG)	-
A8	SW-YOBI(2)	Spare circuit	SW Input (Close to GND)	-
A9	BLOWER_SW	Blower SW Input	SW Input (Close to GND)	A/C blower
A10	AC_BLOWER_REQ	Blower operation request Input	SW Input (Close to GND)	A/C blower
A11	RL_AC_COMP	AC COMP relay activation output	Relay activation output (GND side)	A/C blower
A12	RL_AC_FAN	AC FAN relay activation output	Relay activation output (GND side)	A/C blower
A13	ISS_START_LO	ISS relay activation output (LO side)	Relay activation output (GND side)	ISS
A14	ISS_START_HI	ISS relay activation output (HI side)	Relay activation output (+B side)	ISS
A15	-	-	-	-
A16	BATT1	BCM power	-	-

Pin #	Name	Signal details
A17	ACC	ACC Input
A18	KEY_RD	RS232 ch1 receive data
A19	MODE_CN	For TTI testing
A20	-	-
A21	CAN2_L	Vehicle CAN bus (Lo)
A22	(GND)	BCM GND (spare terminal)
A23	CAN1_L	Vehicle CAN bus (Lo)
A24	SW_YOBI(3)	Spare circuit
A25	SW_YOBI(1)	Spare circuit
A26	KEY_CK	KEY Check Input
A27	SW;DOOR	Door SW Input
A28	SW;MEM_CLEAR	ISS memory clear SW Input
A29	SW;KOUSHO	High altitude operation SW Input
A30	SW;ISS_MAIN	ISS main SW Input
A31	CLUTCH	Clutch fluid pressure SW Input
A32	(BATT1)	BCM power (spare terminal)

Pin #	Name	Signal details	Spec	Function
B1	-	-	-	-
B2	RS232_RD	RS232 ch2 receive data	TIA232F standard	-
B3	TTI_TEST2	RS232 ch2 transmit data	TIA232F standard	-
B4	MODE_CN	TTI testing	TTI testing	-
B5	SW;MANUAL_TRIGGER	SW for manual trigger	SW Input (Close to GND)	-
B6	RS232_CK	RS232 ch2 Check Input	SW Input (Close to GND)	

Pin #	Name	Signal details
B7	TTI-TEST-SG	ACC Input
B8	-	
B9	RS232_SD	RS232 ch2 transmit data
B10	TTI-TEST1	RS232 ch2 receive data
B11	-	-
B12	RS232_SG	RS232 ch2 SGND

2.4 EOL Flash Process and Examination Items

Details of the EOL flash processing and examination items are shown below.

Item	Communication standard	Implementation conditions	Purpose/confirmation details	Effect when executed correctly	Reference
KWP2000 communication start (initialize)	KWP2000	KEY-ON/Engine start	Check if KWP2000 communication is possible	May not be able to communicate with service tools	6.2.1 6.2.2
ECU lock condition check	KWP2000	KEY-ON/Engine start Before VIN flashing	Check that ECU is not locked to flash VIN	Cannot flash VIN if ECU is locked	6.2.3
Flash VIN	KWP2000	KEY-ON Before VIN download	Flash VIN, which is SPS key data.	SPS may not operate properly	6.2.4
Flash EOL data	KWP2000	KEY-ON Before EOL download	Flash EOL data in vehicle in order to get vehicle data records, for self-diagnostics and maintenance warnings	Cannot record data for vehicle. Possibility of mis-diagnosis. Maintenance warnings for model not carried out	6.2.5 6.2.6
Download VIN	KWP2000	KEY-ON/Engine start After VIN flashed	Check if VIN was properly flashed	SPS may not operate properly	7.2.2
Download EOL data	KWP2000	KEY-ON/Engine start After EOL flashed	Check if EOL data was properly flashed	Cannot record data for vehicle. Possibility of mis-diagnosis. Maintenance warnings for model not carried out	7.2.3 7.2.4
ECU lock/check	KWP2000	KEY-ON/Engine start After VIN flashed	Lock ECU so SPS related items cannot be overwritten.	SPS may not operate properly. Also, self-diagnostic may be triggered.	7.2.5 7.2.6
Multi-display check	-	KEY-ON/Engine start After EOL flashed	Operate multi-meter, confirm fuel consumption is displayed correctly.	Fuel consumption meter does not operate	-
Shared info protocol communication start process (initialize)	RS232	KEY-ON/Engine start	Confirm that RS232C (shared information protocol) used for downloading mimamori data works properly.	Mimamori data may now be properly downloaded in the market.	8.4
DTC clear/ download	KWP2000 or RS232	KEY-ON/Engine start After VIN & EOL data flashed. KEY_OFF, wait 15s or more	Clear DTC that occurred due to manufacturing methods before/after EOL data flashing, confirm no DTC raised.	When DTC is detected in post-shipping service, cannot determine if DTC occurred in-market or at factory.	8.5 8.6
Clear all mimamori data	KWP2000 or RS232	KEY-ON/Engine start After VIN & EOL data flashed (before shipping from factory (*))	Clear data that may have been accidentally recorded at the factory.	Data accidentally recorded during factory inspection may remain and after shipping, cannot determine if data was recorded in-market or at factory.	8.7 8.8

(*) Mimamori all clear process is carried out after this process is executed, and at a time/process step when DTC will not be triggered for the factory inspection of other systems.

3. Communication Protocols

3.1. EOL Communication Protocols

The new mimamori ECU EOL data flashing/inspection uses the KEYWORD Protocol 2000 (hereinafter shortened to “KWP2000”).
For details about KWP2000, please refer to the follow ISO regulations:

- ISO/WD 14230-1 — Road Vehicles - Diagnostic Systems - Keyword Protocol 2000 - Physical Layer MAY 1994
- ISO/WD 14230-2 - Road Vehicles - Diagnostic Systems - Keyword Protocol 2000 - Part 2: Data Link Layer FEB 1997
- ISO/WD 14230-3 - Road Vehicles - Diagnostic Systems - Keyword Protocol 2000 - Part3: Application Layer FEB 1997

- Keyword Protocol 2000 Data Link Layer Recommended Practice - Ver. 1.5 OCT 1 1997
- Keyword Protocol 2000 Implementation of Diagnostic Services Recommended Practice - Ver. 1.5 OCT 1 1997

3.2 Specialized Vehicle Equipment/Data Download Communication Protocol

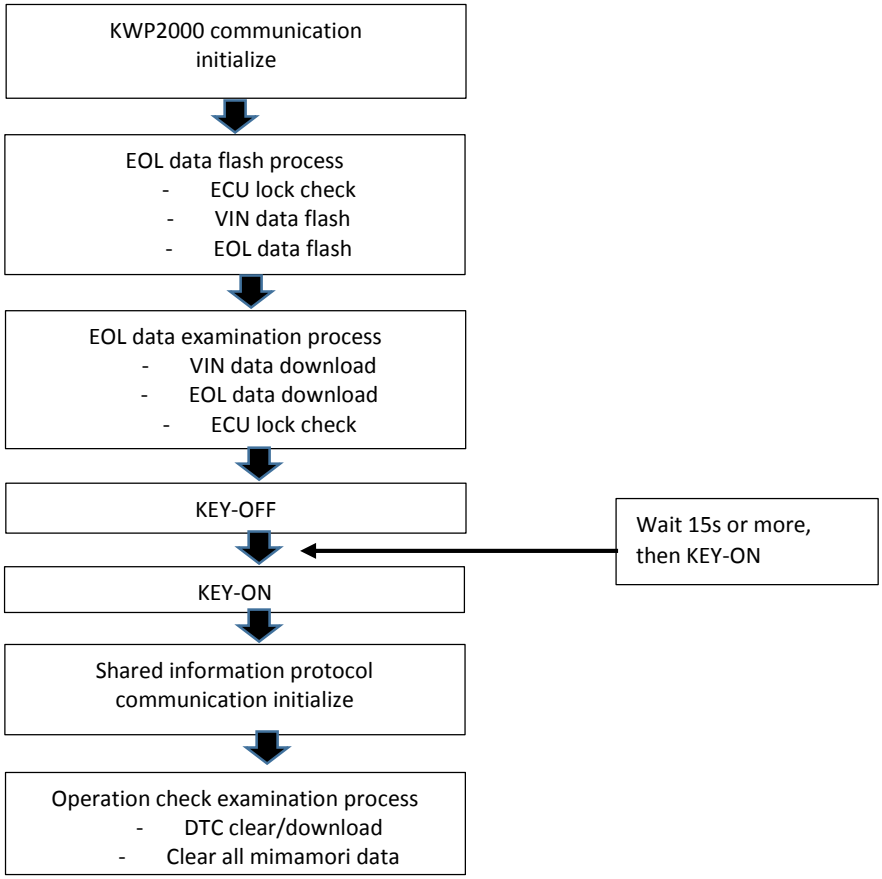
Investigation of whether communication for specialized vehicle equipment/data download can be performed correctly uses the shared information protocol (Isuzu & TTI’s own protocol).
For details, refer to “Shared information vehicle equipment interface specifications document”.

4. Related Requirement Specifications Documents

- New mimamori EOL input data specifications document
- Shared information vehicle equipment interface specifications document

5. EOL Input Process/Examination Flow

The EOL input process/examination flow is shown below:



6. EOL Data Flash Process

6.1 Data for flashing

The mimamori ECU flashes the following data:

Data for mimamori (For details refer to section 8.1)

- Installed ECU data
- Engine model data
- T/M type data
- Tire radius data
- Final gear ration data
- Vehicle model data
- Electrical equipment data
- Gear ratio
- Vehicle speed correction coefficient
- New mimamori calibration data

Maintenance data (For details, refer to section 8.2)

- Engine oil deterioration
- Starter life
- Exhaust brake valve OH
- ACG OH
- T/H OH

SPS support

- VIN

6.2. Flash Process

The flash process is shown below.
The tester (factory equipment) sends each of the request messages in the order shown in this chapter, and checks if the correct acknowledgement message is returned from the ECU.
In the following process, all of the requests use a 3-byte header, but the mimamori ECU can also support 4 byte headers. In addition, during communication, a Mode $3E (Tester Alive) message is transmitted at regular intervals. However, if communication is not interrupted for 5 seconds or more, transmission of the Mode $3E (Tester Alive) message is not necessary.
Furthermore, if a response outside of the acknowledgement response (including negative acknowledgement) is returned, ignoring the following exceptions, the instructions from the tester were not correctly sent to the ECU, and therefore the request is resent.
If communication cannot be achieved even with a repeat of the request, some kind of error has occurred, and a separate response should be carried out.

Exception 1: Security access

For the SEED request from the tester, if an acknowledgement response with a SEED value that is not $0000 is returned, that ECU is already locked for some reason, and the WIN cannot be flashed.
In this case, in order to continue with the EOL process, exchange the ECU or release the security using the Tech-II SPS PC/key.

Exception 2: Tester standby request

If the requested process cannot be completed within a specified time, in order to communicate that the request was correctly received and is being processed, the ECU will send a tester standby request response (response code: $78 negative acknowledgement). If this tester standby request response is received, the unit will wait in accordance with IS14230-2 section 4.4.1.

6.2.1 Start Diagnostic Communication

As a continuation of the wake up pattern defined in ISO 14230-2 section 5.1.4.2.3 Fast Initialization, the following request is sent, and communication is started. If the following acknowledgement response is returned, this step is OK.

Diagnostic communication start	Fmt	Tgt	Src	LEN/SID/LID/Data/CS							Notes
Request (tester→ ECU)	81	80	Fx	81	CS						
Response (ECU→tester)	83	Fx	80	C1	6B	8F	CS				

Fmt: Format Byte
Tgt: Target Address
Src: Source Address
LEN: Length Byte
Fx: Tester Address ($F0~$FD)

SID: service Identifier
LID: Service Identifier
Data: Data Bytes
CS: Check Sum

All abbreviations same below

6.2.2 Diagnostic Session Start (Programming mode)

The following request is sent, and diagnostic session starts in programming mode.
If the following acknowledgement response is returned, this step is OK.

Diagnostic communication start	Fmt	Tgt	Src	LEN/SID/LID/Data/CS							Notes
Request (tester→ ECU)	82	80	Fx	10	85	CS					$85: programming mode
Response (ECU→tester)	82	Fx	80	50	85	CS					

6.2.3 Security Access (Check security conditions)

The following request is sent, and the cancellation of the ECU security (SEED=$0000) is checked.
If the following acknowledgement response is returned, this step is OK.
If the security is operational (SEED≠$0000), use Tech-II SPS PC/key to cancel security or replace ECU.

Security check	Fmt	Tgt	Src	LEN/SID/LID/Data/CS							Notes
Request (tester→ ECU)	82	80	Fx	27	01	CS					Check SEED=$0000
Response (ECU→tester)	84	Fx	80	67	01	00	00	CS			

6.2.4 Flash ECU ID (flash VIN)

The following request is sent, VIN (ECU ID) data is flashed as ASCII code.
If the following acknowledgement response is returned, this step is OK.

VIN flashing	Fmt	Tgt	Src	LEN/SID/LID/Data/CS							Notes
Request (tester→ ECU)	93	80	Fx	3B	90	xx	…	xx	CS		xx…xx: WIN (ASCII) data
Response (ECU→tester)	82	Fx	80	7B	90	CS					

6.2.5 Flash EOL data 1 (flash mimamori data)

The following request is sent, each mimamori EOL data is flashed.
(For details of the flashed EOL data values, refer to section 8 of this document and the new mimamori EOL input data specification document.)
If the following acknowledgement response is returned, this step is OK.

Flash mimamori EOL data 1	Fmt	Tgt	Src	LEN/SID/LID/Data/CS							Notes
Request (tester→ ECU)	83	80	Fx	3B	**	xx	CS				**: Installed ECU data ($20~$3F) xx: not installed/installed (0x00/0x01)
Response (ECU→tester)	82	Fx	80	7B	**	CS					

(NOTE) For Installed ECU, the installed ECU data ($20~3F) value is repeatedly changed.

Flash mimamori EOL data 2	Fmt	Tgt	Src	LEN/SID/LID/Data/CS							Notes
Request (tester→ ECU)	8A	80	Fx	3B	40	xx	…	xx	CS		xx…xx: ENG model data
Response (ECU→tester)	82	Fx	80	7B	40	CS					

Flash mimamori EOL data 3	Fmt	Tgt	Src	LEN/SID/LID/Data/CS							Notes
Request (tester→ ECU)	8A	80	Fx	3B	41	xx	…	xx	CS		xx…xx: T/M model data
Response (ECU→tester)	82	Fx	80	7B	41	CS					

Flash mimamori EOL data 4	Fmt	Tgt	Src	LEN/SID/LID/Data/CS							Notes
Request (tester→ ECU)	84	80	Fx	3B	42	xx	…	xx	CS		Tire radius data
Response (ECU→tester)	82	Fx	80	7B	42	CS					

Flash mimamori EOL data 5	Fmt	Tgt	Src	LEN/SID/LID/Data/CS							Notes
Request (tester→ ECU)	84	80	Fx	3B	43	xx	…	xx	CS		xx…xx: Final gear ratio
Response (ECU→tester)	82	Fx	80	7B	43	CS					

Flash mimamori EOL data 6	Fmt	Tgt	Src	LEN/SID/LID/Data/CS							Notes
Request (tester→ ECU)	83	80	Fx	3B	44	xx	CS				xx: Vehicle model data
Response (ECU→tester)	82	Fx	80	7B	44	CS					

Flash mimamori EOL data 7	Fmt	Tgt	Src	LEN/SID/LID/Data/CS							Notes
Request (tester→ ECU)	83	80	Fx	3B	45	xx	CS				xx: Electrical equipment data
Response (ECU→tester)	82	Fx	80	7B	45	CS					

Flash mimamori EOL data 8	Fmt	Tgt	Src	LEN/SID/LID/Data/CS							Notes
Request (tester→ ECU)	A8	80	Fx	3B	47	xx	…	xx	CS		xx…xx: Gear ratio data
Response (ECU→tester)	82	Fx	80	7B	47	CS					

Flash mimamori EOL data 9	Fmt	Tgt	Src	LEN/SID/LID/Data/CS							Notes
Request (tester→ ECU)	83	80	Fx	3B	48	xx	CS				xx: Vehicle speed correction coefficient data
Response (ECU→tester)	82	Fx	80	7B	48	CS					

Flash mimamori EOL data 10	Fmt	Tgt	Src	LEN/SID/LID/Data/CS							Notes
Request (tester→ ECU)	8A	80	Fx	3B	50	xx	…	xx	CS		xx…xx: New mimamori calibration
Response (ECU→tester)	82	Fx	80	7B	50	CS					

6.2.6 Flash EOL data 2 (flashing data for preventative maintenance)

The following request is sent, and each of the preventative maintenance EOL data items is flashed.
(For details of the flashed EOL data values, refer to section 8 of this document and the new mimamori EOL input data specification document.)
If the following acknowledgement response is returned, this step is OK.

Flash mimamori EOL data 1	Fmt	Tgt	Src	LEN/SID/LID/Data/CS							Notes
Request (tester→ ECU)	86	80	Fx	3B	52	xx	…	xx	CS		xx…xx: Minimum guaranteed drive distance for engine oil life
Response (ECU→tester)	82	Fx	80	7B	52	CS					

Flash mimamori EOL data 2	Fmt	Tgt	Src	LEN/SID/LID/Data/CS							Notes
Request (tester→ ECU)	86	80	Fx	3B	53	xx	…	xx	CS		xx…xx: Drive distance for engine oil life warning
Response (ECU→tester)	82	Fx	80	7B	53	CS					

Flash mimamori EOL data 3	Fmt	Tgt	Src	LEN/SID/LID/Data/CS							Notes
Request (tester→ ECU)	86	80	Fx	3B	54	xx	…	xx	CS		xx…xx: Engine oil life prior warning drive distance
Response (ECU→tester)	82	Fx	80	7B	54	CS					

Flash mimamori EOL data 4	Fmt	Tgt	Src	LEN/SID/LID/Data/CS							Notes
Request (tester→ ECU)	86	80	Fx	3B	55	xx	…	xx	CS		xx…xx: Engine oil life warning total engine rotations
Response (ECU→tester)	82	Fx	80	7B	55	CS					

Flash mimamori EOL data 5	Fmt	Tgt	Src	LEN/SID/LID/Data/CS							Notes
Request (tester→ ECU)	86	80	Fx	3B	56	xx	…	xx	CS		xx…xx: Engine oil life prior warning total rotations
Response (ECU→tester)	82	Fx	80	7B	56	CS					

Flash mimamori EOL data 6	Fmt	Tgt	Src	LEN/SID/LID/Data/CS							Notes
Request (tester→ ECU)	86	80	Fx	3B	57	xx	…	xx	CS		xx…xx: Engine oil life warning insoluble material
Response (ECU→tester)	82	Fx	80	7B	57	CS					

Flash mimamori EOL data 7	Fmt	Tgt	Src	LEN/SID/LID/Data/CS							Notes
Request (tester→ ECU)	86	80	Fx	3B	58	xx	…	xx	CS		xx…xx: Engine oil life prior warning insoluble material
Response (ECU→tester)	82	Fx	80	7B	58	CS					

Flash mimamori EOL data 8	Fmt	Tgt	Src	LEN/SID/LID/Data/CS							Notes
Request (tester→ ECU)	86	80	Fx	3B	59	xx	…	xx	CS		xx…xx: Starter life warning number of starts data
Response (ECU→tester)	82	Fx	80	7B	59	CS					

Flash mimamori EOL data 9	Fmt	Tgt	Src	LEN/SID/LID/Data/CS							Notes
Request (tester→ ECU)	86	80	Fx	3B	5A	xx	…	xx	CS		xx…xx: Exhaust brake valve OH warning number of actuations
Response (ECU→tester)	82	Fx	80	7B	5A	CS					

Flash mimamori EOL data 10	Fmt	Tgt	Src	LEN/SID/LID/Data/CS							Notes
Request (tester→ ECU)	86	80	Fx	3B	5B	xx	…	xx	CS		xx…xx: ACG OH warning drive distance
Response (ECU→tester)	82	Fx	80	7B	5B	CS					

Flash mimamori EOL data 11	Fmt	Tgt	Src	LEN/SID/LID/Data/CS							Notes
Request (tester→ ECU)	86	80	Fx	3B	5C	xx	…	xx	CS		xx…xx: ACG OH warning number of operations
Response (ECU→tester)	82	Fx	80	7B	5C	CS					

Flash mimamori EOL data 12	Fmt	Tgt	Src	LEN/SID/LID/Data/CS							Notes
Request (tester→ ECU)	86	80	Fx	3B	5D	xx	…	xx	CS		xx…xx: T/M OH warning number of gear engagements
Response (ECU→tester)	82	Fx	80	7B	5D	CS					

6.2.7 Complete diagnostic communication

The following request is sent, and communication is completed.
If the following acknowledgement response is returned, this step is OK.

Complete communication	Fmt	Tgt	Src	LEN/SID/LID/Data/CS							Notes
Request (tester→ ECU)	81	80	Fx	82							
Response (ECU→tester)	81	Fx	80	C2							

7. EOL Data Examination Process

7.1. EOL Examination Items

In the examination, the following items are checked:

- The correct ECU ID/EOL data was flashed
- The ECU is locked

7.2 EOL Examination Procedure

The examination procedure is shown below.
The tester (factory equipment) sends each of the request messages in the order shown in this chapter, and checks if the data is correct in the acknowledgement response message returned from the ECU.
In the following process, all of the requests use a 3-byte header, but the mimamori ECU can also support 4 byte headers. In addition, during communication, a Mode $3E (Tester Alive) message is transmitted at regular interval. However, if communication is not interrupted for 5 seconds or more, transmission of the Mode $3E (Tester Alive) message is not necessary.
Furthermore, if a response outside of the acknowledgement response (including negative acknowledgement) is returned, ignoring the following exception, the instructions from the tester were not correctly sent to the ECU, and therefore the request is resent.
If communication cannot be achieved even with a repeat of the request, some kind of error has occurred, and a separate response should be carried out.

Exception: Tester standby request

Similar to the previous section, the tester standby request response (response code: $78 negative acknowledgement) indicates that the request was received correctly by the ECU, but processing could not be completed within the specified time.
If this tester standby request response is received, the unit will wait in accordance with IS14230-2 section 4.4.1.

7.2.1 Start Communication

As a continuation of the wake up pattern defined in ISO 14230-2 section 5.1.4.2.3 Fast Initialization, the following request is sent, and communication is started. If the following acknowledgement response is returned, this step is OK.

Diagnostic communication start	Fmt	Tgt	Src	LEN/SID/LID/Data/CS							Notes
Request (tester→ ECU)	81	80	Fx	81	CS						
Response (ECU→tester)	83	Fx	80	C1	6B	8F	CS				

7.2.2 ECU ID Download (VIN download)

The following request is sent, and the VIN (ECU ID) is downloaded.
If the downloaded VIN and the flashed VIN have the same value, this step is OK.

VIN download	Fmt	Tgt	Src	LEN/SID/LID/Data/CS							Notes
Request (tester→ ECU)	82	80	Fx	1A	90	CS					Check VIN
Response (ECU→tester)	93	Fx	80	5A	90	xx	…	xx	CS		

7.2.3 EOL Data Download 1 (download data for mimamori)

The following requests are sent, and each of the mimamori EOL data is downloaded.
If the downloaded data is the same as the flashed data, this step is OK.

Mimamori EOL data download 1	Fmt	Tgt	Src	LEN/SID/LID/Data/CS							Notes
Request (tester→ ECU)	82	80	Fx	21	**	CS					**: Installed ECU data ($20~$3F)
Response (ECU→tester)	83	Fx	80	61	**	xx	CS				**: not installed/installed (0x00/0x01)

Mimamori EOL data download 2	Fmt	Tgt	Src	LEN/SID/LID/Data/CS							Notes
Request (tester→ ECU)	82	80	Fx	21	40	CS					Installed engine model
Response (ECU→tester)	8A	Fx	80	61	40	xx	..	xx	CS		

Mimamori EOL data download 3	Fmt	Tgt	Src	LEN/SID/LID/Data/CS							Notes
Request (tester→ ECU)	82	80	Fx	21	41	CS					Installed T/M
Response (ECU→tester)	8A	Fx	80	61	41	xx	...	xx	CS		

Mimamori EOL data download 4	Fmt	Tgt	Src	LEN/SID/LID/Data/CS							Notes
Request (tester→ ECU)	82	80	Fx	21	42	CS					Tire radius
Response (ECU→tester)	8A	Fx	80	61	42	xx	...	xx	CS		

Mimamori EOL data download 5	Fmt	Tgt	Src	LEN/SID/LID/Data/CS							Notes
Request (tester→ ECU)	82	80	Fx	21	43	CS					Final ratio
Response (ECU→tester)	8A	Fx	80	61	43	xx	...	xx	CS		

Mimamori EOL data download 6	Fmt	Tgt	Src	LEN/SID/LID/Data/CS							Notes
Request (tester→ ECU)	82	80	Fx	21	44	CS					Vehicle model data
Response (ECU→tester)	83	Fx	80	61	44	xx	CS				

Mimamori EOL data download 7	Fmt	Tgt	Src	LEN/SID/LID/Data/CS							Notes
Request (tester→ ECU)	82	80	Fx	21	45	CS					Electrical equipment data
Response (ECU→tester)	83	Fx	80	61	45	xx	CS				

Mimamori EOL data download 8	Fmt	Tgt	Src	LEN/SID/LID/Data/CS							Notes
Request (tester→ ECU)	82	80	Fx	21	47	CS					Gear ratio
Response (ECU→tester)	A8	Fx	80	61	47	xx	…	xx	CS		

Mimamori EOL data download 9	Fmt	Tgt	Src	LEN/SID/LID/Data/CS							Notes
Request (tester→ ECU)	82	80	Fx	21	48	CS					Vehicle speed correction coefficient
Response (ECU→tester)	83	Fx	80	61	48	xx	CS				

Mimamori EOL data download 10	Fmt	Tgt	Src	LEN/SID/LID/Data/CS							Notes
Request (tester→ ECU)	82	80	Fx	21	50	CS					New mimamori calibration
Response (ECU→tester)	8A	Fx	80	61	50	xx	…	xx	CS		

7.2.4. EOL Data Download 2 (download data for preventative maintenance)

The following request is sent, and each of the EOL data items is downloaded.
If the downloaded data is the same as the flashed data, this step is OK.

Download EOL preventative data 1	Fmt	Tgt	Src	LEN/SID/LID/Data/CS							Notes
Request (tester→ ECU)	82	80	Fx	21	52	CS					Minimum guaranteed drive distance for engine oil life
Response (ECU→tester)	86	Fx	80	61	52	xx	…	xx	CS		

Download EOL preventative data 2	Fmt	Tgt	Src	LEN/SID/LID/Data/CS							Notes
Request (tester→ ECU)	82	80	Fx	21	53	CS					Drive distance for engine oil life warning
Response (ECU→tester)	86	Fx	80	61	53	xx	…	xx	CS		

Download EOL preventative data 3	Fmt	Tgt	Src	LEN/SID/LID/Data/CS							Notes
Request (tester→ ECU)	82	80	Fx	21	54	CS					Engine oil life prior warning drive distance
Response (ECU→tester)	86	Fx	80	61	54	xx	…	xx	CS		

Download EOL preventative data 4	Fmt	Tgt	Src	LEN/SID/LID/Data/CS							Notes
Request (tester→ ECU)	82	80	Fx	21	55	CS					Engine oil life warning total engine revolutions
Response (ECU→tester)	86	Fx	80	61	55	xx	…	xx	CS		

Download EOL preventative data 5	Fmt	Tgt	Src	LEN/SID/LID/Data/CS							Notes
Request (tester→ ECU)	82	80	Fx	3B	56	CS					Engine oil life prior warning total rotations
Response (ECU→tester)	86	Fx	80	7B	56	xx	…	xx	CS		

Download EOL preventative data 6	Fmt	Tgt	Src	LEN/SID/LID/Data/CS							Notes
Request (tester→ ECU)	82	80	Fx	3B	57	CS					Engine oil life warning insoluble material
Response (ECU→tester)	86	Fx	80	7B	57	xx	…	xx	CS		

Download EOL preventative data 7	Fmt	Tgt	Src	LEN/SID/LID/Data/CS							Notes
Request (tester→ ECU)	82	80	Fx	3B	58	CS					Engine oil life prior warning insoluble material
Response (ECU→tester)	86	Fx	80	7B	58	xx	…	xx	CS		

Download EOL preventative data 8	Fmt	Tgt	Src	LEN/SID/LID/Data/CS							Notes
Request (tester→ ECU)	82	80	Fx	3B	59	CS					Starter life warning number of starts
Response (ECU→tester)	86	Fx	80	7B	59	xx	…	xx	CS		

Download EOL preventative data 9	Fmt	Tgt	Src	LEN/SID/LID/Data/CS							Notes
Request (tester→ ECU)	82	80	Fx	3B	5A	CS					Exhaust brake valve OH warning number of actuations
Response (ECU→tester)	86	Fx	80	7B	5A	xx	…	xx	CS		

Download EOL preventative data 10	Fmt	Tgt	Src	LEN/SID/LID/Data/CS							Notes
Request (tester→ ECU)	82	80	Fx	3B	5B	CS					ACG OH warning drive distance
Response (ECU→tester)	86	Fx	80	7B	5B	xx	…	xx	CS		

Download EOL preventative data 11	Fmt	Tgt	Src	LEN/SID/LID/Data/CS							Notes
Request (tester→ ECU)	82	80	Fx	3B	5C	CS					ACG OH warning number of operations
Response (ECU→tester)	86	Fx	80	7B	5C	xx	…	xx	CS		

Download EOL preventative data 12	Fmt	Tgt	Src	LEN/SID/LID/Data/CS							Notes
Request (tester→ ECU)	82	80	Fx	3B	5D	CS					T/M OH warning number of gear engagements
Response (ECU→tester)	86	Fx	80	7B	5D	xx	…	xx	CS		

7.2.5 Security Access (ECU lock)

The following request is sent, ECU security (flash control) is activated.
If the following acknowledgement response is returned, this step is OK.
This process is carried out after the correct flashing of the VIN data has been confirmed.

ECU lock	Fmt	Tgt	Src	LEN/SID/LID/Data/CS							Notes
Request (tester→ ECU)	82	80	Fx	31	08	CS					LID $08: Activate security instruction
Response (ECU→tester)	82	Fx	80	71	08	CS					

7.2.6 Security Access (Check security access)

The following request is sent, and whether the ECU is locked is checked.
If the following acknowledgement response is returned, and SEED in the data is $0000, this step is OK.

Security check	Fmt	Tgt	Src	LEN/SID/LID/Data/CS							Notes
Request (tester→ ECU)	82	80	Fx	27	01	CS					Confirm that SEED(xxxx)≠0000
Response (ECU→tester)	82	Fx	80	67	01	xx	xx	CS			

7.2.7 Complete Diagnostic Communication

The following request is sent, and communication is completed.
If the following acknowledgement response is returned, this step is OK.

Complete communication	Fmt	Tgt	Src	LEN/SID/LID/Data/CS							Notes
Request (tester→ ECU)	81	80	Fx	82	CS						
Response (ECU→tester)	81	Fx	80	C2	CS						4-byte header

8. Operation Confirmation Examination Processing

8.1 Operation Confirmation Preparation

The operation confirmation process is not performed immediately after the EOL data flashing/examination is completed. After the EOL examination has been normally completed, temporarily turn the KEY OFF, and after waiting for 15 seconds or more, turn the KEY ON again, and then carry out the operation confirmation process.
In this examination, the shared information protocol is initialized, the ability to communicate with the new mimamori ECU using the shared information protocol is confirmed, then the DTC clear/check, and the clear all mimamori data processes are carried out.
Note that the DTC clear/check, and the clear all mimamori data processes can be carried out either through RS232C (shared information protocol) or KWP2000 communication.

8.2. Confirmation Items

(1) Start shared information protocol communication
Carry out the start shared information protocol communication (initialize) process, and confirm that communication is possible between the new mimamori ECU using the shared information protocol.

(2) DTC clear/check

When executed via RS232C (shared information protocol)
After executing the DTC clear process, confirm that the DTC is not set.
Confirm that all downloaded statuses are 0x80: no failure.

When executed via KWP2000 communication
Execute the DTC clear process, confirm that positive response is received from ECU.
After executing the DTC clear process, confirm that DTC is not set.

(3) Clear all mimamori data

When executed via RS232C (shared information protocol)
After executing the clear all mimamori data process, wait 4 or more seconds, then confirm that the mimamori frequency data recording region and DRM/EDR recording region have been cleared (downloaded data is all 0).
(Because clearing all of the mimamori data regions requires significant time, wait 4 or more seconds to carry out the confirmation)

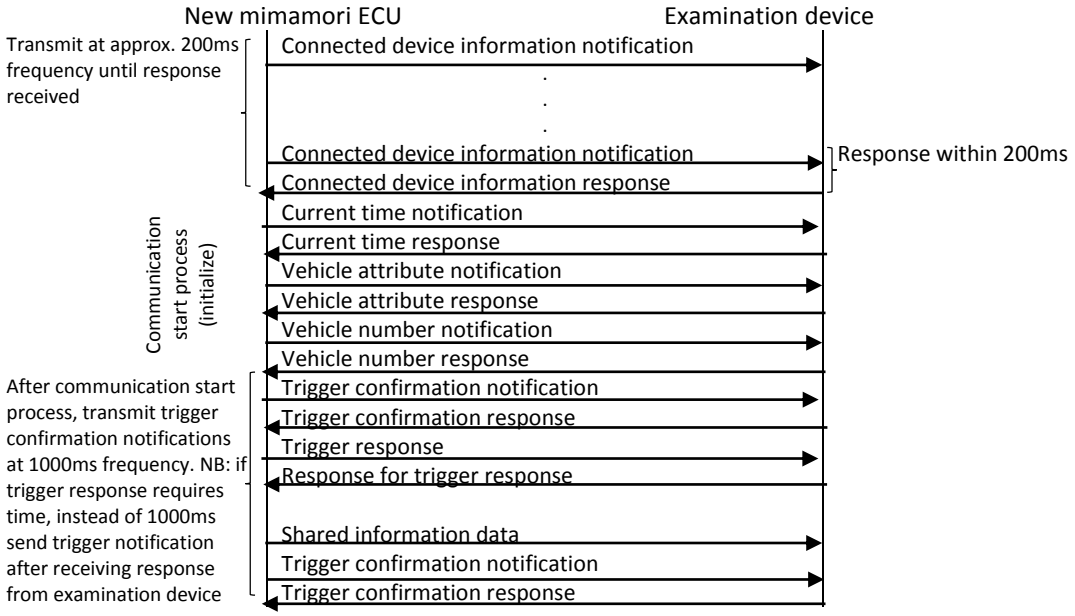
When executed via KWP2000 communication
Execute the clear all mimamori data process, confirm that positive response is received from ECU.

8.3 Confirmation Procedure

The operation confirmation examination procedure is shown below.
The tester (factory equipment) sends each of the request messages in the order shown in this chapter, and checks if the correct data values are in the response message returned from the ECU.

8.4 Shared Information Protocol Communication Start Process

The shared information protocol communication procedure is shown below:



(NOTE) For notifications/responses from the new mimamori ECU, because communication will stop if there is no response for more than 200ms, the examination device will return a response to a new mimamori ECU notification/response within 200ms.
(After communication has been completed once, new communication is carried out from another execution of the communication start process)
In addition, after the communication start process has succeeded, the shared information data shown below is sent regularly from the mimamori ECU, but because the data (notifications) are unrelated to the examination, they are ignored. (No response is OK.)

Shared information data

STX	Compression identifier	Request origin/target	Total data length		Device number				Message ID		Serial number	Message length		No. of data frame bytes		Shared information B (18byte)			Shared information B (18byte)			ETX	CRC	
02	00	17	00	32	00	00	00	00	31	04	xx	00	26	00	24	xx	…	xx	xx	…	xx	03	xx	xx

Connected device information notification (Mimamori → examination device)

STX	Compression identifier	Request origin/target	Total data length		Device number				Message ID		Serial number	Message length		Data											ETX	CRC	
														Related data Y/N	Connected device info												
															Mfr type ID		Software ver				Confirm		Initial				
02	00	17	80	16	00	00	00	00	31	00	xx	00	0A	FE	F0	01	xx	xx	xx	xx	xx	xx	xx	xx	03	xx	xx

xx: arbitrary data (same below)

Software version/confirmation/initial are arbitrary data. They are not related to the examination, and can be ignored.

Connected device information response

STX	Compression identifier	Request origin/target	Total data length		Device number				Message ID		Serial number	Message length		Data									ETX	CRC	
														Status	Mfr type identifier	Year/month/day/hour/minute/second									
02	00	71	00	14	00	00	00	00	31	00	xx	00	08	00	08	00	00	00	00	00	00	03	xx	xx

Current time notification (mimamori → examination device)

STX	Compression identifier	Request origin/target	Total data length		Device number				Message ID		Serial number	Message length		Data						ETX	CRC	
														Year/month/day/hour/minute/second								
02	00	17	80	13	00	00	00	00	31	0F	xx	00	07	xx	xx	xx	xx	xx	xx	03	xx	xx

The data (year/month/day/hour/minute/second) is arbitrary data. Not related to the examination, and can be ignored.

Current time response (Examination device → mimamori)

STX	Compression identifier	Request origin/target	Total data length		Device number				Message ID		Serial number	Message length		Status	ETX	CRC	
02	00	71	00	0D	00	00	00	00	31	0F	xx	00	01	00	03	xx	xx

Vehicle attribute notification (mimamori → examination device)

STX	Compression	Request	Total data length		Device number				Message ID		Serial number	Message length		Data																		ETX	CRC	
														Engine model (8byte)			T/M model (8byte)			Tire radius		Final gear ratio		Dealer area (80byte)			VIN (20byte)						
02	00	17	80	84	00	00	00	00	31	0C	xx	00	78	xx	…	xx	xx	…	xx	xx	xx	xx	xx	xx	…	xx	xx	…	xx	03	xx	xx

The data region is sending the EOL data inside the mimamori ECU, so it is arbitrary.
Not related to examination, therefore can be ignored.

Vehicle attribute response (examination device → mimamori)

STX	Compression identifier	Request origin/target	Total data length		Device number				Message ID		Serial number	Message length		Status	Received byte count		ETX	CRC	
02	00	71	00	0F	00	00	00	00	31	0C	xx	00	03	00	00	78	03	xx	xx

Vehicle number notification (mimamori → examination device)

STX	Compression identifier	Request origin/target	Total data length		Device number				Message ID		Serial number	Message length		No. of data frame bytes		Shared information A (18byte)			Shared information A (18byte)			ETX	CRC	
02	00	17	80	32	00	00	00	00	31	04	xx	00	26	00	24	xx	…	xx	xx	…	xx	03	xx	xx

Shared information A is arbitrary data. Not related to examination, therefore can be ignored.

Vehicle number response (examination device → mimamori)

STX	Compression identifier	Request origin/target	Total data length		Device number				Message ID		Serial number	Message length		Status	Received byte count		ETX	CRC	
02	00	71	00	0F	00	00	00	00	31	04	xx	00	03	00	00	24	03	xx	xx

8.5. DTC Clear/Check (When executed via shared information protocol)

8.5.1 DTC Clear

In this examination, DTC which were triggered in the factory when EOL was not yet set, etc. are cleared. The process is shown below

Trigger confirmation notification (Mimamori → examination device)

STX	Compression identifier	Request origin/target	Total data length		Device number				Message ID		Serial number	Message length		Device identifier	ETX	CRC	
02	00	17	80	0D	00	00	00	00	31	0B	xx	00	01	08	03	xx	xx

Trigger confirmation response (DTC clear request) (Examination device → mimamori)

STX	Compression identifier	Request origin/target	Total data length		Device number				Message ID		Serial number	Message length		Status	Message count	Trigger code	Message #	Command	ETX	CRC	
02	00	71	00	11	00	00	00	00	31	0B	xx	00	05	00	01	10	01	0E	03	xx	xx

8.5.2 DTC Check

In this examination, whether DTC is raised/triggered is checked.
The process is shown below:

Trigger confirmation notification (Mimamori → examination device)

STX	Compression identifier	Request origin/target	Total data length		Device number				Message ID		Serial number	Message length		Device identifier	ETX	CRC	
02	00	17	80	0D	00	00	00	00	31	0B	xx	00	01	08	03	xx	xx

Trigger confirmation response (download address specified data) (examination device → mimamori)

STX	Compression identifier	Request origin/target	Total data length		Device number				Message ID		Serial number	Message length		Status	Message count	Trigger code	Message #	Address				Data count	ETX	CRC	
02	00	71	00	15	00	00	00	00	31	0B	xx	00	09	00	01	03	01	xx	xx	xx	xx	xx	03	xx	xx

For downloaded DTC address and data count, refer to new mimamori EOL input data specification document.

Trigger response (Mimamori → examination device)

STX	Compression identifier	Request origin/target	Total data length		Device number				Message ID		Serial number	Message length		Trigger code	Message #	Data			ETX	CRC	
02	00	17	80	xx	00	00	00	00	31	0A	xx	00	xx	03	01	xx	…	xx	03	xx	xx

Check is OK if all of the received data (DTC status) is 0x80: no failure.
If even 1 data unit is 0xE0: current failure, a separate failure process is carried out.
For the format of the sent data, refer to the new mimamori EOL input data specification document.

Response for trigger response (examination device → mimamori)

STX	Compression identifier	Request origin/target	Total data length		Device number				Message ID		Serial number	Message length		Status	Received byte count	ETX	CRC	
02	00	71	00	0E	00	00	00	00	31	0A	xx	00	02	00	xx	03	xx	xx

8.6. DTC Clear/Check (When executed via KWP2000)

8.6.1 Diagnostic Communication Start

As a continuation of the wake up pattern defined in ISO 14230-2 section 5.1.4.2.3 Fast Initialization, the following request is sent, and communication is started. If the following acknowledgement response is returned, this step is OK.

Diagnostic communication start	Fmt	Tgt	Src	LEN/SID/LID/Data/CS							Notes
Request (tester→ ECU)	81	80	Fx	81	CS						
Response (ECU→tester)	83	Fx	80	C1	6B	8F	CS				

8.6.2 DTC Clear

In this examination, DTC which were triggered in the factory when EOL was not yet set, etc. are cleared.
The following request is sent, any mimamori frequency data, DRM/EDR data recorded during work processed are erased.
If the following acknowledgement response is returned, this step is OK.

Clear all data	Fmt	Tgt	Src	LEN/SID/LID/Data/CS							Notes
Request (tester→ ECU)	82	80	Fx	31	F0	CS					LID $F0: clear all data instruction
Response (ECU→tester)	82	Fx	80	71	F0	CS					

8.6.3 DTC Check

In this examination, whether DTC is raised/triggered is checked. The following request is sent, and no DTC settings in the ECU are checked. If the following acknowledgement response is returned, this step is OK.

DTC download	Fmt	Tgt	Src	LEN/SID/LID/Data/CS							Notes
Request (tester→ ECU)	84	80	Fx	18	00	FF	00	CS			Confirm number of DTC is 0
Response (ECU→tester)	82	Fx	80	58	00	CS					

Note that, if a DTC is set, the following response will be returned, and a separate error/re-examination process is carried out. The number of set DTC is sent in the ND~DS data.

Failure code download	Fmt	Tgt	Src	LEN/SID/LID/Data/CS								Notes
Response (ECU→tester)	80	Fx	90	LEN	58	ND	DH	DL	DS	…	CS	There are set DTC

LEN: Length Byte ND: Number of DTC
DH: DTC High byte DL: DTC Lo byte
DS: DTC Status (DS=80: no failure, A0: past failure, E0: current failure)

8.6.4 Diagnostic Communication Completion

The following request is sent, and communication is completed.
If the following acknowledgement response is returned, this step is OK.

Communication completion	Fmt	Tgt	Src	LEN/SID/LID/Data/CS							Notes
Request (tester→ ECU)	81	80	Fx	82	CS						
Response (ECU→tester)	81	Fx	80	C2	CS						4 byte header

8.7 Clear All Mimamori Data (when executed via shared information protocol)

8.7.1 Clear All Mimamori Data

The following request is sent, any mimamori frequency data, DRM/EDR data recorded during work processed are erased.

Trigger confirmation notification (Mimamori → examination device)

STX	Compression identifier	Request origin/target	Total data length		Device number				Message ID		Serial number	Message length		Device identifier	ETX	CRC	
02	00	17	80	0D	00	00	00	00	31	0B	xx	00	01	08	03	xx	xx

Trigger confirmation response (Clear all mimamori data request) (Examination device → mimamori)

STX	Compression identifier	Request origin/target	Total data length		Device number				Message ID		Serial number	Message length		Status	Message count	Trigger code	Message #	Command	ETX	CRC	
02	00	71	00	11	00	00	00	00	31	0B	xx	00	05	00	01	10	01	0F	03	xx	xx

In the shared information protocol, for the clear all mimamori request from the examination device side, because a positive response is not returned to indicate that it had been correctly executed, use the previous processes shown in 7.7.2 & 7.7.3 to execute the clear.

8.7.2. Frequency Data Check

The process to check whether the frequency data has been cleared or not is shown below.

Because clearing all the regions requires 4 seconds or more, this check is carried out after more than 4 seconds has elapsed after the clear process was executed.

Trigger confirmation notification (Mimamori → examination device)

STX	Compression identifier	Request origin/target	Total data length		Device number				Message ID		Serial number	Message length		Device identifier	ETX	CRC	
02	00	17	80	0D	00	00	00	00	31	0B	xx	00	01	08	03	xx	xx

Trigger confirmation response (download address specified data) (examination device → mimamori)

STX	Compression identifier	Request origin/target	Total data length		Device number				Message ID		Serial number	Message length		Status	Message count	Trigger code	Message #	Address					Data count		ETX	CRC	
02	00	71	00	15	00	00	00	00	31	0B	xx	00	09	00	01	03	01	xx	xx	xx	xx	xx	03		xx	xx

For the address and number of data for the downloaded frequency data, refer to the new mimamori EOL input data specification document.

Trigger response (Mimamori → examination device)

STX	Compression identifier	Request origin/target	Total data length		Device number				Message ID		Serial number	Message length		Trigger code	Message #	Data			ETX	CRC	
02	00	17	80	xx	00	00	00	00	31	0A	xx	00	xx	03	01	xx	…	xx	03	xx	xx

If all of the regions for the data received from mimamori ECU are 0, this check is OK.

If it is not 0, a separate failure process is carried out.

Response for trigger response (examination device → mimamori)

STX	Compression identifier	Request origin/target	Total data length		Device number				Message ID		Serial number	Message length		Status	Received byte count	ETX	CRC	
02	00	71	00	0E	00	00	00	00	31	0A	xx	00	02	00	xx	03	xx	xx

8.7.3 DRM/EDR Data Check

The process for checking whether or not DRM/EDR data is cleared is shown below.
Because clearing all the regions requires 4 seconds or more, this check is carried out after more than 4 seconds has elapsed after the clear process was executed.

[When 1 address is specified in a single message and downloaded]

If 1 address is specified in a single message and downloaded, the following process is repeated for only the number of target systems in each project

Trigger confirmation notification (Mimamori → examination device)

STX	Compression identifier	Request origin/target	Total data length		Device number				Message ID		Serial number	Message length		Device identifier	ETX	CRC	
02	00	17	80	0D	00	00	00	00	31	0B	xx	00	01	08	03	xx	xx

Trigger confirmation response (download address specified data) (examination device → mimamori)

STX	Compression identifier	Request origin/target	Total data length		Device number				Message ID		Serial number	Message length		Status	Message count	Trigger code	Message #	Address				Data count	ETX	CRC	
02	00	71	00	15	00	00	00	00	31	0B	xx	00	09	00	01	03	01	xx	xx	xx	xx	xx	03	xx	xx

For the address and number of data units for the downloaded DRM/EDR data, refer to the new mimamori EOL input data specification document. Set single time download data to 128 bytes or less. If data exceeds 128 bytes, divide into multiple parts and multiple requests.

Trigger response (Mimamori → examination device)

STX	Compression identifier	Request origin/target	Total data length		Device number				Message ID		Serial number	Message length		Trigger code	Message #	Data			ETX	CRC	
02	00	17	80	xx	00	00	00	00	31	0A	xx	00	xx	03	01	xx	…	xx	03	xx	xx

If all of the regions for the data received from mimamori ECU are 0, this check is OK.
If it is not 0, a separate failure process is carried out.

Response for trigger response (examination device → mimamori)

STX	Compression identifier	Request origin/target	Total data length		Device number				Message ID		Serial number	Message length		Status	Received byte count	ETX	CRC	
02	00	71	00	0E	00	00	00	00	31	0A	xx	00	02	00	xx	03	xx	xx

[When multiple addresses are downloaded in a single message]

Trigger confirmation notification (Mimamori → examination device)

STX	Compression identifier	Request origin/target	Total data length		Device number				Message ID		Serial number	Message length		Device identifier	ETX	CRC	
02	00	17	80	0D	00	00	00	00	31	0B	xx	00	01	08	03	xx	xx

Trigger confirmation response (download address specified data) (examination device → mimamori)

STX	Compression identifier	Request origin/target	Total data length		Device number				Message ID		Serial number	Message length		Status	Message count	Trigger code	Message #	Address				Data count	ETX	CRC	
02	00	71	00	15	00	00	00	00	31	0B	xx	00	09	00	01	03	01	xx	xx	xx	xx	xx	03	xx	xx

For the address and number of data units for the downloaded DRM/EDR data, refer to the new mimamori EOL input data specification document.
Set single time download data to 128 bytes or less. If data exceeds 128 bytes, divide into multiple parts and multiple requests.

Trigger response (Mimamori → examination device)

STX	Compression identifier	Request origin/target	Total data length		Device number				Message ID		Serial number	Message length		Trigger code	Message #	Data			ETX	CRC	
02	00	17	80	xx	00	00	00	00	31	0A	xx	00	xx	03	01	xx	...	xx	03	xx	xx

If all of the regions for the data received from mimamori ECU are 0, this check is OK.
If it is not 0, a separate failure process is carried out.

Response for trigger response (examination device → mimamori)

STX	Compression identifier	Request origin/target	Total data length		Device number				Message ID		Serial number	Message length		Status	Received byte count	ETX	CRC	
02	00	71	00	0E	00	00	00	00	31	0A	xx	00	02	00	xx	03	xx	xx

8.8 Clear All Mimamori Data & Check Process (KWP2000 protocol)

8.8.1 Diagnostic Communication Start

As a continuation of the wake up pattern defined in ISO 14230-2 section 5.1.4.2.3 Fast Initialization, the following request is sent, and communication is started. If the following acknowledgement response is returned, this step is OK.

Diagnostic communication start	Fmt	Tgt	Src	LEN/SID/LID/Data/CS							Notes
Request (tester→ ECU)	81	80	Fx	81	CS						
Response (ECU→tester)	83	Fx	80	C1	6B	8F	CS				

8.8.2 Clear All Mimamori Data

The following request is sent, any mimamori frequency data, DRM/EDR data recorded during work processed are erased.
If the following acknowledgement response is returned, this step is OK.
(If the clearing process is executed via KWP2000, and because a positive response is returned to indicate the process was executed correctly from the new mimamori ECU side, and there is no need to check for recording data download and clearing, similar to the shared information protocol)

Clear all data	Fmt	Tgt	Src	LEN/SID/LID/Data/CS							Notes
Request (tester→ ECU)	82	80	Fx	31	F0	CS					LID $F0: clear all data instruction
Response (ECU→tester)	82	Fx	80	71	F0	CS					

8.8.3 Diagnostic Communication Completion

The following request is sent, and communication is completed.
If the following acknowledgement response is returned, this step is OK.

Communication completion	Fmt	Tgt	Src	LEN/SID/LID/Data/CS							Notes
Request (tester→ ECU)	81	80	Fx	82	CS						
Response (ECU→tester)	81	Fx	80	C2	CS						4-byte header

9. EOL Data

EOL data definitions for use in mimamori ECU are shown below.

9.1 EOL Data Definitions for Mimamori Use

9.1.1. Installed ECU Data (LID: $20~$3F)

If the system is installed, value is set to [0x01: Active], if it is not installed, set to [0x00: Inactive]

ID	System name
$20	ECM (Diesel)
$21	ECM (CNG)
$22	SCR
$23	Turbo Controller
$24	Smoother G
$25	NEES1
$26	NEES2
$27	AT
$28	ESC
$29	ABS/ASR/EBS
$2A	Inter-vehicle distance warning ACC/CMS
$2B	Airsus
$2C	Idling Stop & Start
$2D	Driving concentration warning
$2E	Multi-meter
$2F	Lane departure warning

ID	System name
$30	Hybrid System
$31	Fluid retarder
$32	TPMS
$33	ACT
$34	ECON ECU
$35	Reserve
$36	Reserve
$37	Reserve
$38	Reserve
$39	Reserve
$3A	Reserve
$3B	Reserve
$3C	Reserve
$3D	Reserve
$3E	Reserve
$3F	Reserve

9.1.2. Engine Model Data (LID: $40)

The installed engine is specified using 8 byte ASCII data.

ID	Byte	Contents
40	1	Engine model name 1st character (ASCII code / blank filled with 0x20)
	2	Engine model name 2nd character (ASCII code / blank filled with 0x20)
	3	Engine model name 3rd character (ASCII code / blank filled with 0x20)
	4	Engine model name 4th character (ASCII code / blank filled with 0x20)
	5	Engine model name 5th character (ASCII code / blank filled with 0x20)
	6	Engine model name 6th character (ASCII code / blank filled with 0x20)
	7	Engine model name 7th character (ASCII code / blank filled with 0x20)
	8	Engine model name 8th character (ASCII code / blank filled with 0x20)

9.1.3. T/M Model Data (LID:$41)

The installed engine is specified using 8 byte ASCII data.

ID	Byte	Contents
41	1	T/M model name 1st character (ASCII code / blank filled with 0x20)
	2	T/M model name 2nd character (ASCII code / blank filled with 0x20)
	3	T/M model name 3rd character (ASCII code / blank filled with 0x20)
	4	T/M model name 4th character (ASCII code / blank filled with 0x20)
	5	T/M model name 5th character (ASCII code / blank filled with 0x20)
	6	T/M model name 6th character (ASCII code / blank filled with 0x20)
	7	T/M model name 7th character (ASCII code / blank filled with 0x20)
	8	T/M model name 8th character (ASCII code / blank filled with 0x20)

9.1.4. Tire Radius Data (LID: $42)

The dynamic load radius of the rear wheel is specified as 2 byte UNM (unsigned numeric) data (1bit=1mm).

Example: 380[mm] → 0x017C

9.1.5. Final Ratio Data (LID: $43)

The final drive ratio is specified as 2 byte UNM (unsigned numeric) data (1bit=0.001)

Example: 4.300 → 4300=0x10CC

9.1.6. Vehicle Model Data (LID: $44)

The vehicle model identifier is specified as 1-byte bitmap data.

ID	Bit	Contents	Meaning of 1	Meaning of 0
44	8	Large model (C&E)	Applicable	Not applicable
	7	Mid-size (F)	Applicable	Not applicable
	6	Small model (N)	Applicable	Not applicable
	5	Bus	Applicable	Not applicable
	4	SKW	Applicable	Not applicable
	3	US	Applicable	Not applicable
	2	-	-	-
	1	EXP	Applicable	Not applicable

9.1.7. Electrical Equipment Data (LID: $45)

The electrical equipment identifier is specified as 1-byte bitmap data.

ID	Bit	Contents	Meaning of 1	Meaning of 0
45	8	Air conditioner control	Enable	Disable
	7	-	-	-
	6	-	-	-
	5	-	-	-
	4	-	-	-
	3	-	-	-
	2	CAN communication (J1939)	O1(B): 250k 10(B): 500k 00(B), 11(B): CAN communication/diagnostic stop	
	1			

9.1.8. Gear Ratio Data (LID: $47)

The number of main gear changer steps is specified as 1 byte UNM (unsigned numeric value) data (1bit = 1 step), and the transmission gear ratio is specified as 2 byte UNM (unsigned numeric value) (1bit=0.001), for a total 38 bytes.

ID	Byte	Details
47	1	Input number of main gear changer steps (1bit/step)
	2	Reserve (0x00 fixed)
	3,4	Gear 1: Multiply by 1000 and input as hexadecimal (For unused gear regions, enter 0x00)
	5,6	Gear 2: Multiply by 1000 and input as hexadecimal (For unused gear regions, enter 0x00)
	. . .	. . .
	35, 36	Gear 17: Multiply by 1000 and input as hexadecimal (For unused gear regions, enter 0x00)
	37,38	Gear 18: Multiply by 1000 and input as hexadecimal (For unused gear regions, enter 0x00)

9.1.9. Vehicle Speed Correction Coefficient (LID: $48)

Identify the vehicle speed correction coefficient as 1 byte UNM (unsigned numeric value) data (1bit=1%)

Example: 100[%] → 0x64

9.1.10 New Mimamori Calibration Data (LID: $50)

Specify the new mimamori calibration as 8-byte bitmap data.

ID	byte	bit	Details	Meaning of 1	Meaning of 0
50	1	8	EOL transmission	Enable	Disable
		7	-	-	-
		6	-	-	-
		5	-	-	-
		4	-	-	-
		3	-	-	-
		2	-	-	-
		1	-	-	-
	2-8	-	Reserve	-	-

9.2 Preventative Maintenance EOL Data Definitions

9.2.1. Minimum Guaranteed Drive Distance for Engine Oil Life (LID: $52)

Specify the minimum guaranteed drive distance for engine oil life as 4 byte UNM (unsigned numeric value) data (1bit=1m).

Example: 5,000[km] → 5,000,000[m] → 0x004C4B40

9.2.2 Drive Distance for Engine Oil Life Warning (LID: $53)

Specify the drive distance for engine oil life as 4 byte UNM (unsigned numeric value) data (1bit=1m).

Example: 5,000[km] → 5,000,000[m] → 0x004C4B40

9.2.3 Engine Oil Life Prior Warning Drive Distance (LID: $54)

Specify the engine oil life prior warning drive distance as 4 byte UNM (unsigned numeric value) data (1bit=1m).

Example: 5,000[km] → 5,000,000[m] → 0x004C4B40

9.2.4. Engine Oil Life Warning Total Engine Revolutions (LID: $55)

Specify the engine oil life warning total engine revolutions as 4 byte UNM (unsigned numeric value) data (1bit=1 revolution).

Example: 100,0000,000 [revolution] → 0x05F5E100

9.2.5. Engine Oil Life Prior Warning Total Revolutions (LID: $56)

Specify the engine oil life warning total engine revolutions as 4 byte UNM (unsigned numeric value) data (1bit=1 revolution).

Example: 100,0000,000 [revolution] → 0x05F5E100

9.2.6. Engine Oil Life Warning - Insoluble Material (LID: $57)

Specify the engine oil life warning for insoluble material as 4 byte UNM (unsigned numeric value) data (1bit=1wt%/100h)).

Example: 100,0000,000 [wt%/100h] → 0x05F5E100

9.2.7. Engine Oil Life Prior Warning - Insoluble Material (LID: $58)

Specify the engine oil life prior warning insoluble material as 4 byte UNM (unsigned numeric value) data (1bit=1wt%/100h)).

Example: 100,0000,000 [wt%/100h] → 0x05F5E100

9.2.8. Starter Life Warning Number of Starts Data (LID: $59)

Specify the starter life warning number of starts data as 4 byte UNM (unsigned numeric value) data (1bit=1 activation).

Example: 100,000 [activations] →0x000186A0

9.2.9 Exhaust Brake Valve OH Warning Number of Actuations (LID: $5A)

Specify the exhaust brake valve OH warning number of actuations as 4 byte UNM (unsigned numeric value) data (1bit=1 actuation)

Example: 100,000 [activations] →0x000186A0

9.2.10 ACG OH Warning Drive Distance (LID:$5B)

Specify the ACG OH warning drive distance as 4 byte UNM (unsigned numeric value) data (1bit=1m)

Example: 5,000[km] → 5,000,000[m] → 0x004C4B40

9.2.11 ACG OH Warning Number of Operations (LID:$5C)

Specify the ACG OH warning number of operations as 4 byte UNM (unsigned numeric value) data (1bit=1 actuation)

Example: 100,000 [activations] →0x000186A0

9.2.12 T/M OH Warning Number of Gear Engagements (LID:$5D)

Specify the T/M OH warning number of gear engagements 4 byte UNM (unsigned numeric value) data (1bit=1 gear engagement)

Example: 100,000 [gear engagement] →0x000186A0

10. DTC Codes

The self-diagnostics executed by the mimamori ECU are shown below

DTC	Details	Examination request
P0601	Program flash fault	-
P0602	Security unlock status (ECU lock process not executed)	○
B1000	ECU fault (RAM flash error)	○
B1001	EOL data fault (EOL data not set)	○
B1008	T/M EOL data fault	○
B1011	CAN EOL data fault	○
U2101	Installed ECU EOL data fault	○
U0073	CAN (ISO-CAN) communication failure	○
U2100	CAN (SAE-J1939-CAN) communication failure	○
U2105	Communication failure with ECM	○
U2106	Communication failure with smoother/TCM/ACT	○
U2108	Communication failure to ABS/ASR/EBS/IESC	○
U2112	Communication failure to air suspension ECU	○
U2116	Communication failure to multi-meter	○
U2117	Communication failure to driving concentration warning (DSA)	○
U2144	Communication failure to driving distance warning/ACC/CMS	○
U2202	Communication failure to SCR(DCU)	○
U2203	Communication failure to Turbo Controller	○
U2204	Communication failure to Idling Stop & Start	○
U2205	Communication failure to ECON ECU	○